



ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : Bachelor in Information Technology (Honours) (Internet of Things)
Course Code & Name : IIB43203 - CLOUD COMPUTING
Course Lecturer(s) : Mohd Asyraf Abdul Rahman
Submission Title : GROUP PROJECT
Deadline : Day 7 Month 6 Year 2026 Time 11.59PM
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Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
	MUHAMMAD ASYRAF WAJDI BIN MUSTAFFA AHMAD SYAZWAN BIN SHAHRUDDIN	52227124873 52227124582

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MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY

MARCH 2026 SEMESTER

CLOUD COMPUTING (IIB43203)

GROUP PROJECT

PREPARED BY:

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SUBMISSION ON DATE:
7TH JUNE 2026

A. Amazon S3 Storage Setup

1. Bucket setup

1.1 Open AWS console and choose S3, then click create bucket

Default encryption [info](#)
Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [info](#)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the Storage tab of the [Amazon S3 pricing page](#).

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

► **Advanced settings**

1 After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

[Cancel](#) [Create bucket](#)

1.2 Choose General purpose for General configuration, and set the bucket name to student-registration-system. Click Create Bucket.

Create bucket [info](#)
Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [info](#)

- ☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- ☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

- ☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.
- ☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [info](#)
student-registration-asywan

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

2. Uploading Media

2.1 After bucket successfully created, select the new bucket and click upload.

Successfully created bucket "student-registration-asywan"

student-registration-asywan [info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (0) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.				

[Upload](#)

2.2 Add file of the image and click Upload.

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 total, 12.7 KB)

All files and folders in this table will be uploaded.

Find by name

☐	Name	Folder	Type	Size
☐	Student Thumb up.jpeg	-	image/jpeg	12.7 KB

Destination Info

Destination
[s3://student-registration-asywan](#)

► **Destination details**
Bucket settings that impact new objects stored in the specified destination.

► **Permissions**
Grant public access and access to other AWS accounts.

► **Properties**
Specify storage class, encryption settings, tags, and more.

[Cancel](#) [Upload](#)

Upload succeeded
For more information, see the [Files and folders](#) table.

Upload: status [Close](#)

After you navigate away from this page, the following information is no longer available.

Summary

Destination s3://student-registration-asywan	Succeeded 1 file, 12.7 KB (100.00%)	Failed 0 files, 0 B (0%)
--	---	------------------------------------

[Files and folders](#)

[Configuration](#)

Files and folders (1 total, 12.7 KB)

Find by name

Name	Folder	Type	Size	Status	Error
Student Thumb up.jpeg	-	image/jpeg	12.7 KB	Succeeded	-

B. AWS EC2 setup

1. Launch instance

1.1 in AWS console select EC2, then click Launch instance.

EC2

[Dashboard](#)
[AWS Global View](#)
[Events](#)
▼ **Instances**
[Instances](#)
[Instance Types](#)
[Launch Templates](#)
[Spot Requests](#)
[Savings Plans](#)
[Reserved Instances](#)
[Dedicated Hosts](#)
[Capacity Reservations](#)
[Capacity Manager](#) New

Resources

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	1	Auto Scaling Groups
Dedicated Hosts	0	Elastic IPs
Key pairs	2	Load balancers
Security groups	10	Snapshots

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#) [Migrate a server](#)

1.2 Write the name and tags

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

student-registration-server-asywan

[Add additional tags](#)

1.3 For the OS, choose Ubuntu.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux macOS **Ubuntu** Windows Red Hat SUSE Linux Debian

aws Mac Ubuntu Microsoft Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type Free tier eligible

ami-05cf1e9f73fbad2e2 (64-bit (x86)) / ami-0953e222326856ce (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture	AMI ID	Publish Date	Username	
64-bit (x86)	ami-05cf1e9f73fbad2e2	2026-04-24	ubuntu	Verified provider

1.4 On Create key pair, write the key pair name and choose RSA key pair type. Keep the private key file format in .pem. Click Create key pair.

Create key pair ✕

Key pair name

Key pairs allow you to connect to your instance securely.

studentkey

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

1.5 For network setting, choose Create security group on Firewall (security groups). Tick Allow SSH and Allow HTTP. Make sure to Click Launch instance.

The screenshot shows the AWS Management Console 'Network settings' page. Under 'Firewall (security groups)', the 'Create security group' option is selected. Below, it states 'We'll create a new security group called 'launch-wizard-2' with the following rules:'. Two rules are checked: 'Allow SSH traffic from' (source: Anywhere, 0.0.0.0/0) and 'Allow HTTP traffic from the Internet' (source: Anywhere, 0.0.0.0/0). The 'Summary' panel on the right shows 1 instance, Canonical Ubuntu 24.04 AMI, t3.micro instance type, and 1 volume (8 GiB). The 'Launch instance' button is visible.

2. Setup Apache and default page

2.1 open terminal by select our EC2 instance and click connect

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security
Asyraf-ec2	i-069c361b4e1aae2f4	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-100-52-193-247.co...	100.52.193.247	-	-	disabled	ec2-rds
student-regist...	i-07a6e02a7246facaf	Running	t3.micro	3/3 checks passed	us-east-1d	ec2-3-81-151-210.com...	3.81.151.210	-	-	disabled	ec2-rds

2.2 install Apache

```
ubuntu@ip-172-31-32-124:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

2.3 start and enable Apache

```
ubuntu@ip-172-31-32-124:~$ sudo systemctl start apache2
ubuntu@ip-172-31-32-124:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

2.4 Edit the web page html

```
ubuntu@ip-172-31-32-124:~$ cd /var/www/html
ubuntu@ip-172-31-32-124:/var/www/html$ ls
index.html
ubuntu@ip-172-31-32-124:/var/www/html$ sudo rm index.html
ubuntu@ip-172-31-32-124:/var/www/html$ sudo nano index.html
```

2.5 Put the html code for the web page.

```
GNU nano 7.2
<!DOCTYPE html>
<html>
<head>
<title>Student Registration System</title>
</head>
<body>
<h1>Student Registration System</h1>
<p>Welcome to my AWS Project.</p>
</body>
</html>
```

3. Install PHP and setup registration form

3.1 install the PHP in the terminal and edit the register.php

```
ubuntu@ip-172-31-32-124:/var/www/html$ cd /var/www/html
ubuntu@ip-172-31-32-124:/var/www/html$ sudo nano register.php
```

3.2 Put the code of the registration form

```
GNU nano 7.2
<!DOCTYPE html>

<html>
<head>
    <title>Student Registration System</title>
</head>
<body>

<h1>Student Registration System</h1>

<form method="post">
...
<label>Full Name:</label><br>
<input type="text" name="fullname" required><br><br>

<label>Email Address:</label><br>
<input type="email" name="email" required><br><br>

<label>Phone Number:</label><br>
<input type="text" name="phone" required><br><br>

<input type="submit" value="Register">
...

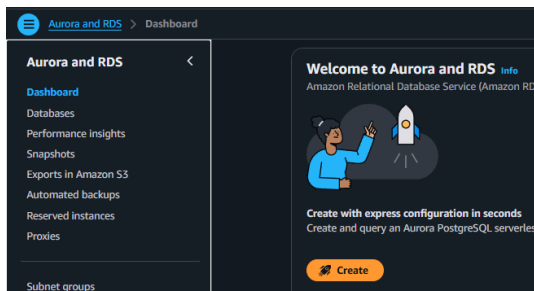
</form>

</body>
</html>
```

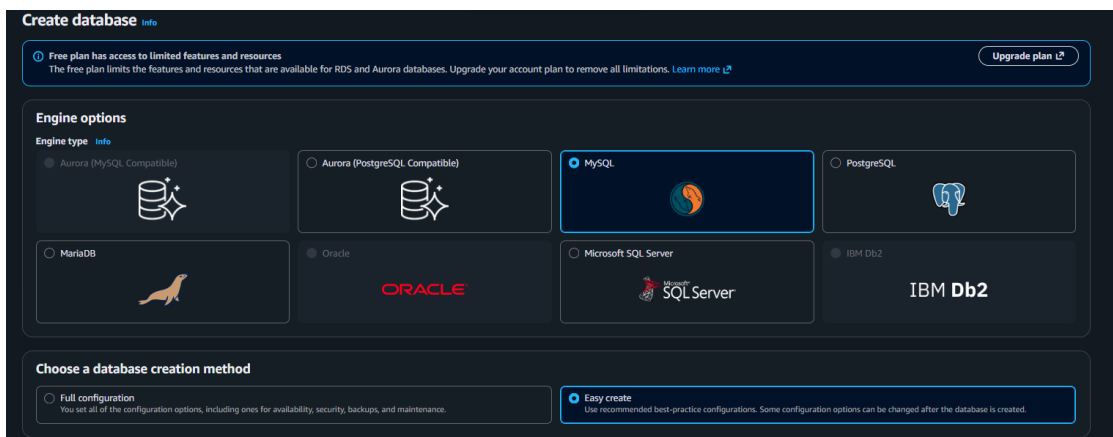
C. Amazon RDS Database Setup

1. Manage the database

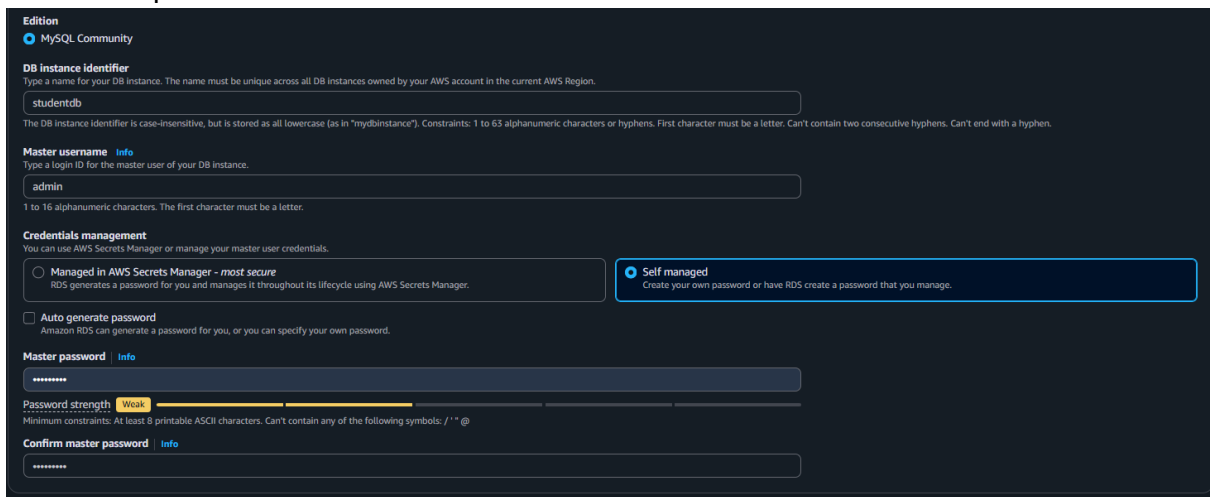
1.1 Click Create



1.2 Choose MySQL for Engine options and choose Easy create for database creation method.



1.3 Write the studentdb in DB instance identifier. Choose Self managed and create the Master password.



1.4 Select Connect to an EC2 compute resource to connect to our EC2 instance earlier. Click Create database.

▼ Set up EC2 connection - optional
You can also set up a connection to an EC2 instance after creating the database. Go to the database list page or the database details page, choose Actions, and then choose Set up to EC2 connection.

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

EC2 instance Info
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-07a6e02a7246facaf
student-registration-server-asywan

ⓘ You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel Create database

2. Connect EC2 to MySQL

2.1 Open the terminal of EC2 and install MySQL

```
ubuntu@ip-172-31-32-124:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-32-124:~$ sudo apt install mysql-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mysql-client is already the newest version (8.0.46-0ubuntu0.24.04.2).
0 upgraded, 0 newly installed, 0 to remove and 40 not upgraded.
```

2.2 Open MySQL prompt and create database table.

```
ubuntu@ip-172-31-32-124:~$ mysql -h studentdb.ce3kimoio74.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 44
Server version: 8.4.8 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE student_registration;
Query OK, 1 row affected (0.03 sec)

mysql> USE student_registration;
Database changed
mysql> CREATE TABLE students (
->   id INT AUTO_INCREMENT PRIMARY KEY,
->   fullname VARCHAR(100),
->   email VARCHAR(100),
->   phone VARCHAR(20)
-> );
Query OK, 0 rows affected (0.12 sec)
```

2.3 Edit register.php again and put the code to connect the website with database.

```
GNU nano 7.2
<?php
$host = "studentdb.ce3kimoi0r74.us-east-1.rds.amazonaws.com";
$user = "admin";
$password = "Admin123!";
$database = "student_registration";

$conn = new mysqli($host, $user, $password, $database);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$message = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $fullname = $_POST['fullname'];
    $email = $_POST['email'];
    $phone = $_POST['phone'];

    $sql = "INSERT INTO students (fullname, email, phone)
            VALUES ('$fullname', '$email', '$phone')";

    if ($conn->query($sql) === TRUE) {
        $message = "Student registered successfully!";
    } else {
        $message = "Error: " . $conn->error;
    }
}

?>
```

```

<!DOCTYPE html>

<html>
<head>
    <title>Student Registration System</title>
</head>
<body>

<h1>Student Registration System</h1>

<p><?php echo $message; ?></p>

<form method="post">
    ...
<label>Full Name:</label><br>
<input type="text" name="fullname" required><br><br>

<label>Email Address:</label><br>
<input type="email" name="email" required><br><br>

<label>Phone Number:</label><br>
<input type="text" name="phone" required><br><br>

<input type="submit" value="Register">
    ...

</form>

</body>
</html>

```

D. Web test

1. Web registration

1.1 fill the form

Student Registration System

Full Name:

Email Address:

Phone Number:

1.2 data registered

Student Registration System

Student registered successfully!

Full Name:

Email Address:

Phone Number:

Register

2. Check the data in terminal

```
mysql> SELECT * FROM students;
+----+-----+-----+-----+
| id | fullname | email | phone |
+----+-----+-----+-----+
| 1 | syazwan | student@unikl | 0123456789 |
+----+-----+-----+-----+
1 row in set (0.00 sec)
```